

Digital Signal Processing (EEE F434/ ECE F434)

Lab 8: DSP-HDL: Operations

Instructions:

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
 2. All the tasks in each lab have to be in one main file, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
 3. Plots should include x label, y label, legend, and titles
 4. File name: ID_Lab8.rar or zip **and** ID_Lab8.pdf
 5. If the report/code is found to be copied, zero marks will be awarded [**Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**]
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1. Write a Verilog code for the Clock generation of 100 MHz, and a 10 ns period.
2. A) Implement a parameterized FIR filter in Verilog (with 4 taps) for the two configurations as below
 - i. Low-Pass: {1, 2, 2, 1}
 - ii. High-Pass: {-1, -2, 2, 1}

B) Implement a first-order IIR LPF in Verilog, which is described by a difference equation $y[n] = 0.5 x[n] + 0.5 y[n-1]$

C) In your test bench, apply the input signal as $x[n] = \{10, 20, 30, 40, -40, -30, -20, -10, 0, 0, 0, 0\}$. Simulate and compare outputs of FIR-LPF, FIR-HPF, and IIR-LPF as designed in questions 1, 2, and 3. Comment on the results obtained?

Note: Generate and observe waveforms in GTKWave.

3. Write a Verilog code for an 8-bit sequence taken in the lecture, $x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$, to compute the DIT-FFT.

Note that you need to design the FFT code such that the input and output are both in the usual order [no bit reversing at the input and output]

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Appendix: Test bench for filter Q2

```
module tb_filters;
  wire clk;
  reg signed [7:0] x_in;
  wire signed [15:0] y_fir_lpf;
  wire signed [15:0] y_fir_hpf;
  wire signed [15:0] y_iir;

  clock_gen cg(.clk(clk));

  fir_filter fir_lpf (.clk(clk), .x_in(x_in), .y_out(y_fir_lpf));

  fir_filter #(
    .H0(-8'sd1), .H1(-8'sd2), .H2(8'sd2), .H3(8'sd1)
  ) fir_hpf (.clk(clk), .x_in(x_in), .y_out(y_fir_hpf));

  iir_filter iir_lp (.clk(clk), .x_in(x_in), .y_out(y_iir));

  initial begin
    $dumpfile("filters_wave.vcd");
    $dumpvars(0, tb_filters);
    $monitor("Time=%0t ps | x_in=%d | FIR_LPF=%d | FIR_HPF=%d | IIR_LPF=%d",
      $time, x_in, y_fir_lpf, y_fir_hpf, y_iir);
  end

  initial begin
    x_in = 0;
    @(posedge clk) x_in <= 10; @(posedge clk) x_in <= 20;
    @(posedge clk) x_in <= 30; @(posedge clk) x_in <= 40;
    @(posedge clk) x_in <= -40; @(posedge clk) x_in <= -30;
    @(posedge clk) x_in <= -20; @(posedge clk) x_in <= -10;
    @(posedge clk) x_in <= 0; @(posedge clk) x_in <= 0;
    @(posedge clk) x_in <= 0; @(posedge clk) x_in <= 0;
    #50; $finish;
  end
end
endmodule
```